

Roll No.

Total Pages : 3

CMCA/M-22 24535
BLOCKCHAIN TECHNOLOGY

Paper–MCA-20-45

Option–(iii)

Time Allowed : 3 Hours]

[Maximum Marks : 75

Note : Attempt **five** questions in all, selecting **one** question from each Unit. Question No. **1** is compulsory. All questions carry equal marks.

Compulsory Question

1. Answer the following questions in brief : 5×3=15
- (a) What is coinbase transaction? Explain.
 - (b) Explain historical evolution of Blockchain in brief.
 - (c) What are Smart Contracts? How are these useful?
 - (d) What is Blockchain mining? Explain in brief.
 - (e) Discuss bitcoin digital keys and addresses.

UNIT–I

2. (a) What is Blockchain? Explain full ecosystem of a Blockchain. 8
- (b) What are smart contracts? Explain working of smart contracts. 7

3. (a) Explain different types of Blockchain with their relative advantages and disadvantages. 8
- (b) Explain different platforms for decentralization in brief. 7

UNIT–II

4. (a) Differentiate between Blockchain and shared database. 7
- (b) Explain the Architecture Blockchain. Also discuss life usecases of Blockchain. 8
5. (a) What is Cryptocurrency? How is it different from fiat currency? Also explain different types of cryptocurrency. 8
- (b) What do you mean by Bitcoin? Explain bitcoin digital keys and addresses in detail. 7

UNIT–III

6. (a) What is double spending? How does Blockchain prevent double-spending? 8
- (b) What is bitcoin network? How payments are made on a bitcoin network? Explain. 7
7. Write short notes on the following : 5×3=15
- (a) Bitcoin wallet.
 - (b) Proof-of-work.
 - (c) Innovations in Bitcoin.

UNIT-IV

8. (a) What is Hyperledger platform? Explain key features of Hyperledger. 8
- (b) Compare and contrast Multichain and Bigchain platforms. 7
9. (a) How can you design a new Blockchain? Explain. 7
- (b) What is dApp? Explain the procedure to design a dApp. 8